OPENFLIGHT FIELD REPORT · 2026-11-01 · INDOOR STUDIO · IWR6843 SELF-TRIGGER

# Studio Session 2026-11-01

OpenFlight on the IWR6843 self-trigger, hitting into a net beside a TrackMan 4. 127 IWR triggers over four runs, matched shot by shot against 76 TrackMan shots exported with SwingSync.

## TL;DR

- **The OPS243 dropped from 30 to 20 kS/s at 11:37:53 and read 1.49 × TrackMan's ball speed until a restart at 12:35.** Its own trigger offset went from 102 to 154 ms and its J3 window from 34 to 48 ms; UART noise sending S2 is the leading suspect. Outside that period, ball speed matched TrackMan to 0.995 ± 0.028.
- **The self-trigger caught 57 of 76 real shots on time (75 %).** Six fired on the takeaway or backswing, ten never fired, three fired 60-70 ms too early.
- **False triggers are frequent and SNR does not fix them.** Addresses, waggles, practice swings, placing a ball and walking up all fired it at SNR 1.5, 2.0 and 3.0; each one blinds the IWR for 7.7 s and can cost the next real shot.
- **IWR launch was 2.4° MAE (n=8)**, three times the project's 0.83° TrackMan baseline. `--iwr6843-net-m` must be unit → net; entering ball → net (3.0 m) clipped every track.
- **Real-time scheduling buys nothing:** S! reaches the OPS in 0.17 ± 0.01 ms either way.

**57 / 76**
real shots captured on time

**×1.49**
OPS ball speed, 11:37:53-12:34 (27 shots)

**61**
false shots and no-ball triggers with no TrackMan shot

**2.4°**
IWR launch MAE, against a 0.83° baseline

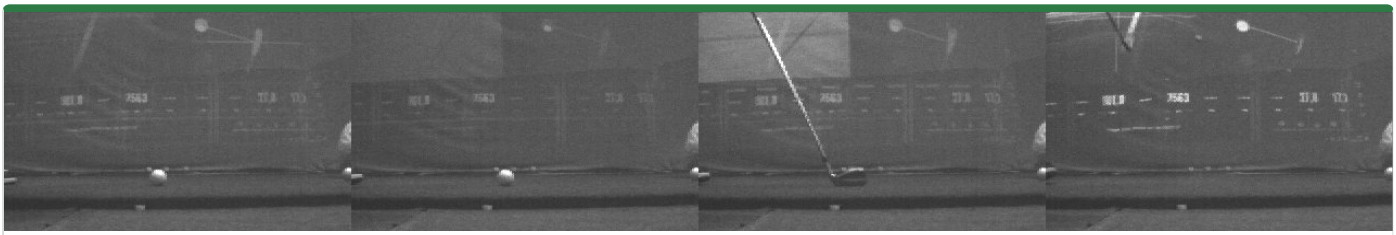
**0.17 ms**
S! → OPS, normal or real-time

# Lessons learned

1. **The OPS changed its sampling mid-run.** At 11:37:53, with no host command in between, ball speed went to 1.486 ± 0.035 × TrackMan, club speed to 1.37 ×, the J3 post-trigger window from ~32 to ~48 ms, the log's `post-trigger` from 34.5 to 0.0 ms, and the clock sync from 36 to 59 valid readings. The OPS's own trigger offset in every capture went from exactly 102.00 to exactly 154.00 ms: 3072 pre-trigger samples at 30 kS/s, then at 20 kS/s. It was not a reboot (the OPS powers up at 30 kS/s in continuous mode) and not the move (that came at ~11:52). The OPS reported UART receive errors (4 noise, 4 framing, 1 break since boot); one candidate is a re-arm `S#24` that lost its # and read as `S2` (20 kS/s), to be confirmed against OmniPreSense's API. A restart fixed it.
2. **`--iwr6843-net-m` is unit → net.** 3.0 m (meant as ball → net) left the tracker 0.5 m, about 10 ms of flight. Replaying the same dumps with 4.6 m tracked 24-41 ms on every real shot. Unit → net was 5.30 m.
3. **The self-trigger caught 75 % of real shots on time.** Early triggers on the takeaway left the OPS reading the slow backswing (20-50 mph) and the real shot fell in the blind window.
4. **False triggers are frequent; SNR does not fix them.** 20 saved shots and 41 no-ball triggers had no TrackMan shot behind them. At SNR 3.0 a practice swing still triggered and real shots went missing.
5. **Every trigger blinds the IWR for ~7.7 s** with `--debug`: 7.1 s readback plus a radar restart.
6. **IWR launch angles were 3× worse than the baseline**: 2.4° MAE, 1.4° median, +1.5° bias on 8 clean shots, with the radar 0.04 m above the floor, lower than any validated mount (0.1524 m). Replaying 31 shots with the ball centre at its real 0.061 m (4 cm mat + 2.1 cm radius) instead of 0.04 m cut the bias on the 23 good tracks from +0.87° to +0.39°, but 8 shots stayed 5-11° off: wrong tracks, not geometry, dominate the error.
7. **Measure both heights from the floor the unit stands on.** Radar height to the centre of the IWR antenna group, ball height to the centre of the ball (mat + 0.021 m).
8. **The clocks were fine.** NTP (systemd-timesyncd) syncs whenever there is a connection, and the RTC battery charges and kept time across power-off. TrackMan's clock, not the Pi's, was 425.5 s off.
9. **Real-time scheduling buys nothing.** The window jitter that looked halved was the 11:37:53 switch.
10. **"TI radar capture failed" is misleading.** It shows on every shot because the firmware never has an onboard result ready; the data is still saved.
11. **Launch OpenFlight from the desktop icon.** Over SSH with `nohup … &` the server ignored SIGINT, and SIGTERM skipped the OPS idle step.

# What the camera saw

Each strip shows four frames: 89 ms and 22 ms before the trigger, the trigger frame, and 44 ms after. OV9281 at 320×200, 150 ms before and 50 ms after each IWR trigger.



CAPTURED

## Real shot, captured on time

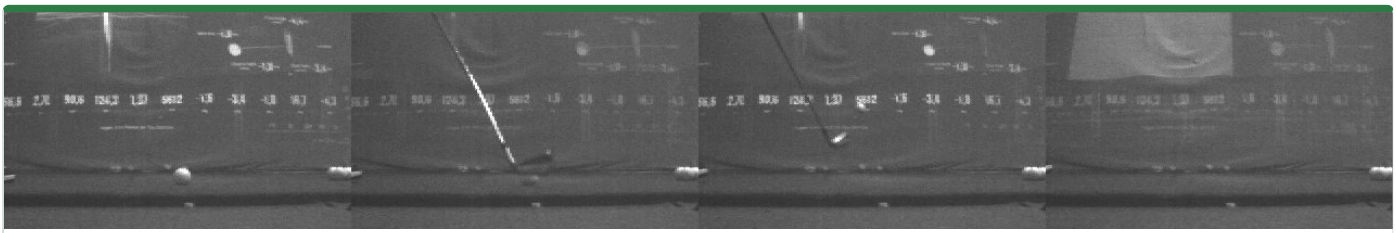
11:32:50 · OpenFlight 120.5 mph, TrackMan 116.4 mph · IWR launch 23.3°, TrackMan 21.3°



WRONG SPEED

## Real shot, OPS reading ×1.49

12:03:20 · OpenFlight 192.0 mph, TrackMan 123.3 mph · inside the 11:37:53-12:34 period



CAPTURED

## Real shot after the restart

12:49:39 · SNR 3.0 · OpenFlight 122.2 mph, TrackMan 118.8 mph



LOST OR FALSE

## Trigger on the takeaway

12:00:52 · fired 0.79 s before TrackMan's impact · saved as 20.4 mph, the real 114.8 mph shot was lost



LOST OR FALSE

## Real shot, trigger 70 ms early

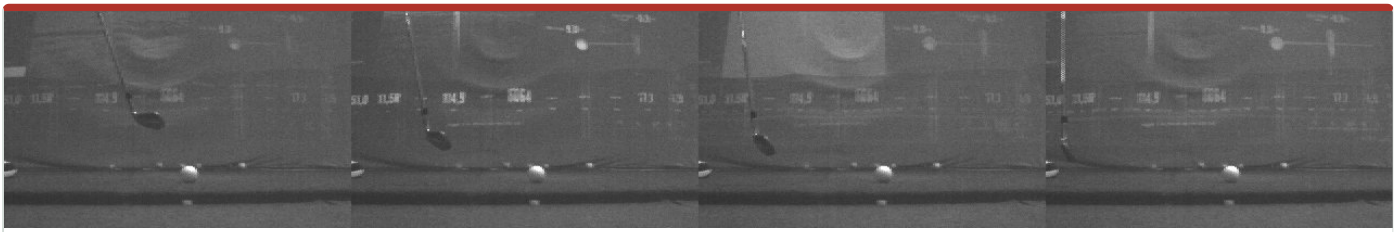
12:48:19 · both the OPS and the camera window closed before impact · TrackMan 121.9 mph, OpenFlight nothing



LOST OR FALSE

## Address

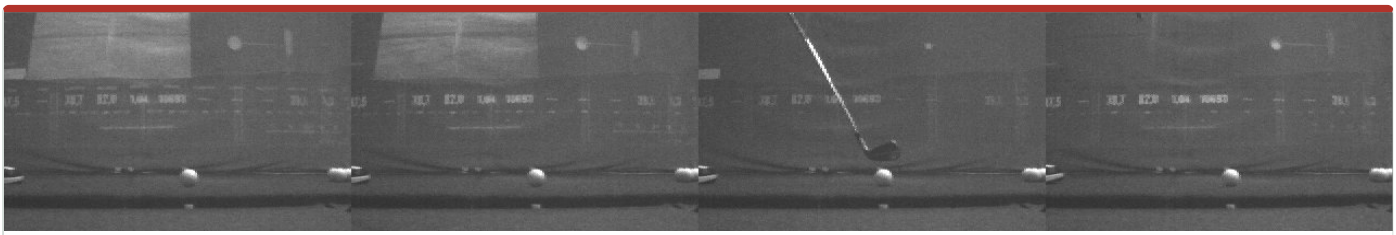
12:15:10 · club resting behind the ball · saved as false shot #36, 20.3 mph from 3 readings



LOST OR FALSE

## Practice swing

12:02:40 · ball never moves · saved as false shot #26, 22.5 mph from 1 reading



LOST OR FALSE

## Practice swing at SNR 3.0

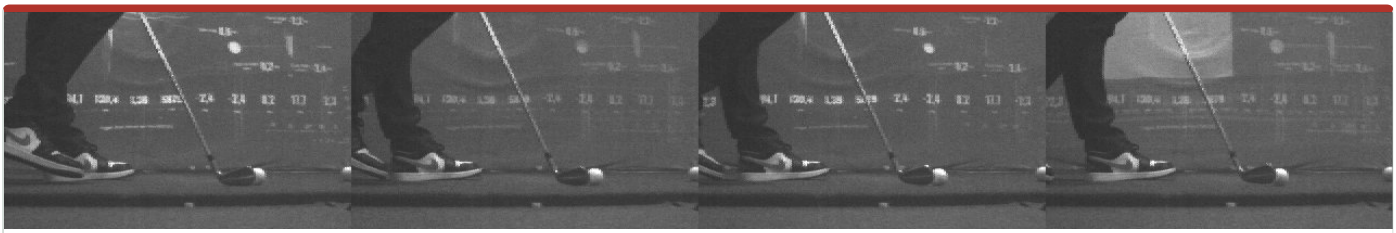
12:48:01 · saved as false shot #2, a 74.0 mph “ball” from 29 readings



LOST OR FALSE

## Walking up to the screen

12:04:43 · saved as false shot #30, 27.0 mph from 4 readings



LOST OR FALSE

## Placing a ball

12:53:22 · OPS saw no ball, so no shot, but the IWR was blind for 7.7 s

# Every capture

All 127 camera captures, each matched to its IWR trigger, the OPS verdict and TrackMan. Speeds are ball/club in mph; the last column is the OPS capture window from J3 pin 1.

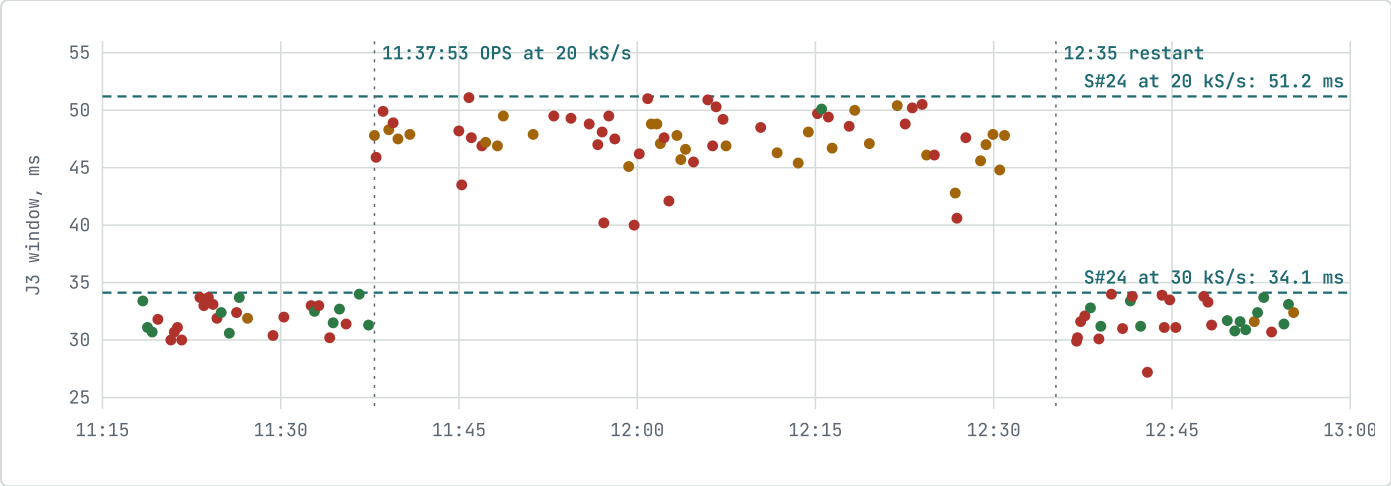
| TIME | SNR | OUTCOME | OPENFLIGHT | TRACKMAN | J3 MS |
|---|---|---|---|---|---|
| 11:18:24 | 1.5 | REAL SHOT, CAPTURED | #1 113.6/79.9 | #1 116.0/82.6 | 33.4 |
| 11:18:47 | 1.5 | REAL SHOT, CAPTURED | #2 111.1/80.3 | #2 113.5/83.8 | 31.1 |
| 11:19:11 | 1.5 | REAL SHOT, CAPTURED | #3 113.0/81.8 | #3 115.1/84.2 | 30.7 |
| 11:19:40 | 1.5 | FALSE SHOT | #4 18.3/16.0 | - | 31.8 |
| 11:20:45 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 30.0 |
| 11:21:02 | 1.5 | FALSE SHOT | #5 44.1/30.4 | - | 30.7 |
| 11:21:19 | 1.5 | NO-BALL TRIGGER | no ball (1) | - | 31.1 |
| 11:21:40 | 1.5 | FALSE SHOT | #6 16.7/- | - | 30.0 |
| 11:23:12 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 33.7 |
| 11:23:31 | 1.5 | REAL SHOT, OPS SAW NO BALL | no ball (0) | #4 16.9/17.1 | 33.0 |
| 11:23:54 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 33.7 |
| 11:24:18 | 1.5 | NO-BALL TRIGGER | no ball (2) | - | 33.1 |
| 11:24:38 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 31.9 |
| 11:24:59 | 1.5 | REAL SHOT, CAPTURED | #7 113.7/81.9 | #5 115.9/84.8 | 32.4 |
| 11:25:40 | 1.5 | REAL SHOT, CAPTURED | #8 113.2/82.2 | #6 115.1/82.9 | 30.6 |
| 11:26:17 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 32.4 |
| 11:26:30 | 1.5 | REAL SHOT, CAPTURED | #9 110.8/80.2 | #7 112.4/82.9 | 33.7 |
| 11:27:13 | 1.5 | REAL SHOT, WRONG OPS SPEED | #10 71.5/40.9 | #8 101.0/79.4 | 31.9 |
| 11:29:21 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 30.4 |
| 11:30:16 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 32.0 |
| 11:32:33 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 33.0 |
| 11:32:50 | 1.5 | REAL SHOT, CAPTURED | #1 120.5/88.7 | #9 116.4/83.6 | 32.5 |
| 11:33:12 | 1.5 | FALSE SHOT | #2 20.6/17.0 | - | 33.0 |
| 11:34:07 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 30.2 |
| 11:34:24 | 1.5 | REAL SHOT, CAPTURED | #3 113.4/84.4 | #10 115.2/82.3 | 31.5 |
| 11:34:56 | 1.5 | REAL SHOT, CAPTURED | #4 116.5/79.3 | #11 120.1/88.6 | 32.7 |
| 11:35:30 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 31.4 |
| 11:36:36 | 1.5 | REAL SHOT, CAPTURED | #5 79.2/76.0 | #13 82.0/77.1 | 34.0 |
| 11:37:23 | 1.5 | REAL SHOT, CAPTURED | #6 83.5/80.3 | #14 86.7/78.0 | 31.3 |
| 11:37:53 | 1.5 | REAL SHOT, SPEED X1.49 | #7 135.8/125.7 | #15 86.3/77.9 | 47.8 |
| 11:38:01 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 45.9 |
| 11:38:36 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 49.9 |
| 11:39:05 | 1.5 | REAL SHOT, SPEED X1.49 | #8 118.8/116.0 | #16 82.8/76.4 | 48.3 |
| 11:39:27 | 1.5 | EARLY TRIGGER, REAL SHOT LOST | #9 55.4/37.8 | #17 77.7/67.9 | 48.9 |
| 11:39:51 | 1.5 | REAL SHOT, SPEED X1.49 | #10 128.5/125.4 | #18 89.5/76.2 | 47.5 |
| 11:40:52 | 1.5 | REAL SHOT, SPEED X1.49 | #11 194.4/85.1 | #19 124.3/84.8 | 47.9 |
| 11:44:59 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 48.2 |
| 11:45:13 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 43.5 |
| 11:45:50 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 51.1 |
| 11:46:02 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 47.6 |
| 11:46:56 | 1.5 | EARLY TRIGGER, REAL SHOT LOST | #12 24.8/- | #20 128.3/87.2 | 46.9 |
| 11:47:15 | 1.5 | REAL SHOT, SPEED X1.49 | #13 199.1/116.2 | #21 130.0/88.1 | 47.2 |

| TIME | SNR | OUTCOME | OPENFLIGHT | TRACKMAN | J3 MS |
|---|---|---|---|---|---|
| 11:48:14 | 1.5 | REAL SHOT, SPEED X1.49 | #14 200.3/130.6 | #22 130.2/87.5 | 46.9 |
| 11:48:43 | 1.5 | REAL SHOT, SPEED X1.49 | #15 189.7/116.1 | #23 128.1/86.9 | 49.5 |
| 11:51:14 | 1.5 | REAL SHOT, SPEED X1.49 | #16 157.7/109.8 | #25 107.3/77.9 | 47.9 |
| 11:52:58 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 49.5 |
| 11:54:25 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 49.3 |
| 11:55:57 | 1.5 | FALSE SHOT | #17 20.2/16.9 | - | 48.8 |
| 11:56:40 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 47.0 |
| 11:57:03 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 48.1 |
| 11:57:11 | 1.5 | FALSE SHOT | #18 187.2/117.3 | - | 40.2 |
| 11:57:36 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 49.5 |
| 11:58:07 | 1.5 | FALSE SHOT | #19 19.7/16.7 | - | 47.5 |
| 11:59:16 | 1.5 | REAL SHOT, SPEED X1.49 | #20 181.0/114.4 | #26 123.2/84.4 | 45.1 |
| 11:59:44 | 1.5 | EARLY TRIGGER, REAL SHOT LOST | #21 50.9/34.2 | #27 126.4/85.9 | 40.0 |
| 12:00:10 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 46.2 |
| 12:00:52 | 1.5 | EARLY TRIGGER, REAL SHOT LOST | #22 20.4/16.8 | #28 114.8/81.0 | 51.0 |
| 12:01:11 | 1.5 | REAL SHOT, WRONG OPS SPEED | #23 22.8/17.5 | #29 117.3/83.1 | 48.8 |
| 12:01:39 | 1.5 | REAL SHOT, SPEED X1.49 | #24 186.3/111.5 | #30 126.0/85.7 | 48.8 |
| 12:01:56 | 1.5 | REAL SHOT, SPEED X1.49 | #25 180.3/115.1 | #31 122.1/88.2 | 47.1 |
| 12:02:15 | 1.5 | EARLY TRIGGER, REAL SHOT LOST | no ball (7) | #32 114.9/82.0 | 47.6 |
| 12:02:40 | 1.5 | FALSE SHOT | #26 22.5/- | - | 42.1 |
| 12:03:20 | 1.5 | REAL SHOT, SPEED X1.49 | #27 192.0/117.2 | #33 123.3/83.4 | 47.8 |
| 12:03:40 | 1.5 | REAL SHOT, SPEED X1.49 | #28 170.9/108.1 | #34 115.9/87.2 | 45.7 |
| 12:04:03 | 1.5 | REAL SHOT, SPEED X1.49 | #29 187.8/124.6 | #35 126.9/86.1 | 46.6 |
| 12:04:43 | 1.5 | FALSE SHOT | #30 27.0/- | - | 45.5 |
| 12:05:56 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 50.9 |
| 12:06:20 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 46.9 |
| 12:06:38 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 50.3 |
| 12:07:13 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 49.2 |
| 12:07:28 | 1.5 | REAL SHOT, WRONG OPS SPEED | #31 27.4/20.7 | #36 121.7/83.3 | 46.9 |
| 12:10:23 | 1.5 | EARLY TRIGGER, REAL SHOT LOST | #32 49.5/41.7 | #37 126.7/87.4 | 48.5 |
| 12:11:46 | 1.5 | REAL SHOT, SPEED X1.49 | #33 180.6/118.8 | #38 122.9/85.6 | 46.3 |
| 12:13:33 | 1.5 | REAL SHOT, SPEED X1.49 | #34 208.2/114.1 | #40 140.7/93.1 | 45.4 |
| 12:14:23 | 1.5 | REAL SHOT, SPEED X1.49 | #35 177.0/112.4 | #41 119.3/80.7 | 48.1 |
| 12:15:10 | 1.5 | FALSE SHOT | #36 20.3/- | - | 49.7 |
| 12:15:30 | 1.5 | REAL SHOT, CAPTURED | #37 132.8/84.0 | #42 131.6/86.5 | 50.1 |
| 12:16:05 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 49.4 |
| 12:16:24 | 1.5 | REAL SHOT, SPEED X1.49 | #38 165.0/106.6 | #43 112.5/81.4 | 46.7 |
| 12:17:50 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 48.6 |
| 12:18:18 | 1.5 | REAL SHOT, SPEED X1.49 | #39 164.2/114.2 | #44 110.9/81.3 | 50.0 |
| 12:19:32 | 1.5 | REAL SHOT, SPEED X1.49 | #40 169.3/101.8 | #45 115.2/81.6 | 47.1 |
| 12:21:53 | 1.5 | REAL SHOT, SPEED X1.49 | #41 163.8/102.1 | #46 111.4/82.0 | 50.4 |
| 12:22:33 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 48.8 |
| 12:23:08 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 50.2 |

| TIME | SNR | OUTCOME | OPENFLIGHT | TRACKMAN | J3 MS |
|---|---|---|---|---|---|
| 12:23:58 | 1.5 | REAL SHOT, OPS SAW NO BALL | no ball (0) | #48 108.4/80.4 | 50.5 |
| 12:24:21 | 1.5 | REAL SHOT, SPEED X1.49 | #42 160.4/106.1 | #49 109.3/80.1 | 46.1 |
| 12:25:00 | 1.5 | FALSE SHOT | #43 66.4/44.5 | - | 46.1 |
| 12:26:45 | 1.5 | REAL SHOT, SPEED X1.49 | #44 164.9/114.6 | #51 112.1/82.5 | 42.8 |
| 12:26:53 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 40.6 |
| 12:27:38 | 1.5 | NO-BALL TRIGGER | no ball (0) | - | 47.6 |
| 12:28:53 | 1.5 | REAL SHOT, SPEED X1.49 | #45 166.0/118.5 | #53 112.9/81.5 | 45.6 |
| 12:29:20 | 1.5 | REAL SHOT, SPEED X1.49 | #46 158.2/111.5 | #54 107.3/82.6 | 47.0 |
| 12:29:56 | 1.5 | REAL SHOT, WRONG OPS SPEED | #47 24.3/16.4 | #55 107.8/81.5 | 47.9 |
| 12:30:30 | 1.5 | REAL SHOT, SPEED X1.49 | #48 145.6/99.8 | #57 99.2/77.6 | 44.8 |
| 12:30:54 | 1.5 | REAL SHOT, SPEED X1.49 | #49 159.6/117.0 | #58 108.3/80.6 | 47.8 |
| 12:36:58 | 2.0 | NO-BALL TRIGGER | no ball (0) | - | 29.9 |
| 12:37:03 | 2.0 | NO-BALL TRIGGER | no ball (0) | - | 30.2 |
| 12:37:18 | 2.0 | FALSE SHOT | #1 16.7/- | - | 31.6 |
| 12:37:39 | 2.0 | NO-BALL TRIGGER | no ball (0) | - | 32.1 |
| 12:38:08 | 2.0 | REAL SHOT, CAPTURED | #2 118.7/79.7 | #59 113.6/81.5 | 32.8 |
| 12:38:50 | 2.0 | NO-BALL TRIGGER | no ball (0) | - | 30.1 |
| 12:39:01 | 2.0 | REAL SHOT, CAPTURED | #3 117.8/108.7 | #60 112.5/82.1 | 31.2 |
| 12:39:53 | 2.0 | NO-BALL TRIGGER | no ball (0) | - | 34.0 |
| 12:40:50 | 2.0 | FALSE SHOT | #4 55.4/43.8 | - | 31.0 |
| 12:41:29 | 2.0 | REAL SHOT, CAPTURED | #5 63.8/54.4 | #62 67.1/54.2 | 33.4 |
| 12:41:40 | 2.0 | FALSE SHOT | #6 54.0/36.7 | - | 33.8 |
| 12:42:21 | 2.0 | REAL SHOT, CAPTURED | #7 84.7/75.1 | #63 82.0/78.7 | 31.2 |
| 12:42:55 | 2.0 | FALSE SHOT | #8 57.9/41.0 | - | 27.2 |
| 12:44:10 | 2.0 | FALSE SHOT | #9 27.1/- | - | 33.9 |
| 12:44:21 | 2.0 | FALSE SHOT | #10 31.6/22.4 | - | 31.1 |
| 12:44:48 | 2.0 | FALSE SHOT | #11 49.3/- | - | 33.5 |
| 12:45:19 | 2.0 | NO-BALL TRIGGER | no ball (0) | - | 31.1 |
| 12:47:40 | 3.0 | FALSE SHOT | #1 31.0/21.3 | - | 33.8 |
| 12:48:01 | 3.0 | FALSE SHOT | #2 74.0/60.0 | - | 33.3 |
| 12:48:19 | 3.0 | REAL SHOT, OPS SAW NO BALL | no ball (0) | #64 121.9/83.7 | 31.3 |
| 12:49:39 | 3.0 | REAL SHOT, CAPTURED | #3 122.2/79.8 | #66 118.8/82.4 | 31.7 |
| 12:50:16 | 3.0 | REAL SHOT, CAPTURED | #4 116.1/80.7 | #67 116.6/82.4 | 30.8 |
| 12:50:43 | 3.0 | REAL SHOT, CAPTURED | #5 79.4/59.4 | #68 79.9/57.8 | 31.6 |
| 12:51:11 | 3.0 | REAL SHOT, CAPTURED | #6 75.2/68.2 | #69 72.6/52.0 | 30.9 |
| 12:51:55 | 3.0 | REAL SHOT, WRONG OPS SPEED | #7 29.9/- | #71 48.9/34.4 | 31.6 |
| 12:52:12 | 3.0 | REAL SHOT, CAPTURED | #8 126.0/84.2 | #72 127.9/87.2 | 32.4 |
| 12:52:43 | 3.0 | REAL SHOT, CAPTURED | #9 128.9/84.1 | #73 130.4/94.1 | 33.7 |
| 12:53:22 | 3.0 | NO-BALL TRIGGER | no ball (0) | - | 30.7 |
| 12:54:25 | 3.0 | REAL SHOT, CAPTURED | #10 127.2/84.1 | #74 129.5/93.6 | 31.4 |
| 12:54:48 | 3.0 | REAL SHOT, CAPTURED | #11 133.0/88.7 | #75 135.5/91.4 | 33.1 |
| 12:55:13 | 3.0 | REAL SHOT, WRONG OPS SPEED | #12 18.8/- | #76 131.2/89.9 | 32.4 |

# J3 pin 1 on GPIO18

The OPS243 raises J3 pin 1 when it receives S! and drops it when its post-trigger capture closes. Each dot is one trigger. At 11:37:53 the window jumps from the 34 ms that S#24 takes at 30 kS/s to the ~48-51 ms it takes at 20 kS/s, and drops back after the restart at 12:35.



# Against TrackMan

TrackMan's clock ran 425.5 s ahead of the Pi's NTP time; at that offset 66 of its 76 shots line up with an IWR trigger within 1 s. TrackMan stamps a shot 70 ± 29 ms after OpenFlight's trigger, so a trigger more than 0.3 s before TrackMan's stamp is an early one.

## Did each real shot get captured?

| PERIOD | TRACKMAN | ON TIME | EARLY | NO TRIGGER | ON TIME, NO BALL |
|---|---|---|---|---|---|
| SNR 1.5, net 3.0 · 11:15-11:31 | 8 | 7 | 0 | 0 | 1 |
| SNR 1.5 · 11:32-11:37 | 6 | 5 | 0 | 1 | 0 |
| SNR 1.5, OPS ×1.49 · 11:37-12:34 | 44 | 31 | 6 | 6 | 1 |
| SNR 2.0 · 12:35-12:45 | 5 | 4 | 0 | 1 | 0 |
| SNR 3.0 · 12:45-12:55 | 13 | 10 | 0 | 2 | 1 |
| **Total** | **76** | **57** | **6** | **10** | **3** |

Six on-time captures still got a wrong OPS speed (0.14-0.71 × TrackMan). OpenFlight shots with no TrackMan shot behind them: 11 at SNR 1.5, 7 at SNR 2.0, 2 at SNR 3.0.

## Accuracy against the project's validation

| MEASURE | THIS SESSION | PROJECT BASELINE |
|---|---|---|
| OPS ball speed − TrackMan, before cosine correction | −0.5 mph bias, 2.6 mph MAE (n=24) | −2.15 mph bias, 2.83 mph MAE (128 shots) |
| OPS club speed − TrackMan | −0.4 mph bias, 4.9 mph MAE, p50 2.7 (n=24) | — |
| IWR LCMF launch − TrackMan, irons | +1.5° bias, 2.4° MAE, p50 1.4° (n=8) | −0.04° bias, 0.83° MAE, p50 0.67° (87 shots) |
| LCMF coverage on captured shots | 24 of 57 (42 %) | 87 % strict, irons and wedges |

Baseline from OpenFlight's launch-angle report and `speed_correction.py`. LCMF takes OPS speed as an input, so the 16 results inside the ×1.49 period or the net-3.0 run (2.6° MAE) are not a fair test.

### The ×1.49 period

| | BEFORE 11:37:53, AFTER 12:35 | 11:37:53-12:34 |
|---|---|---|
| OPS / TrackMan ball speed | 0.995 ± 0.028 (n=24) | 1.486 ± 0.035 (n=27) |
| OPS / TrackMan club speed | 1.00 ± 0.11 | 1.37 ± 0.13 |
| J3 window (S#24 post-trigger) | 30-34 ms | 46-51 ms |
| Log `post-trigger` | 34.5 ms | 0.0 ms |

It started between two shots with nothing logged but a normal rearm, and survived the move next to the TrackMan at ~11:52, so it is not TrackMan proximity.

## Runs

| START | TIME | SNR | NET | TRIGGERS | SHOTS | NO-BALL |
|---|---|---|---|---|---|---|
| 111546 | 11:15-11:31 | 1.5 | 3.0 m ✗ | 20 | 10 | 10 |
| 113221 | 11:32-12:34 | 1.5 | 5.30 m | 76 | 49 | 27 |
| 123514 | 12:35-12:45 | 2.0 | 5.30 m | 17 | 11 | 6 |
| 124549 | 12:45-12:55 | 3.0 | 5.30 m | 14 | 12 | 2 |

“No-ball” is the UI's “impact detected” without metrics. Real-time scheduling was on from 12:13:44 to 12:34.

## Net distance

Offline replay of the IWR ball tracker on the dumps from 11:18-11:27, which ran with `--iwr6843-net-m 3.0`:

| DUMP | NET 3.0 | NET 4.6 | NO NET |
|---|---|---|---|
| `111824_001` | reject, 9 ms | ok, 27 ms | ok, 27 ms |
| `111911_003` | reject, 9 ms | ok, 41 ms | ok, 43 ms |
| `112540_015` | reject, 13 ms | ok, 41 ms | ok, 41 ms |
| `112630_017` | ok, 16 ms | ok, 27 ms | ok, 27 ms |

# Trigger latency and the real-time A/B

OPS J3 pin 1 was wired to the Pi's GPIO18 and logged with `gpiomon`; each camera capture records the host's trigger time on the same monotonic clock.

| WINDOW | SCHEDULING | TRIGGERS | S! → OPS | OPS WINDOW |
|---|---|---|---|---|
| 11:32-12:13 | normal | 54 | 0.17 ± 0.01 ms | 44.8 ± 6.1 ms (30-51) |
| 12:13-12:35 | `chrt -a -f 50` | 22 | 0.17 ± 0.01 ms | 47.5 ± 2.5 ms (41-51) |
| 12:35-12:45 | normal, new process | 17 | 0.16 ± 0.01 ms | 31.7 ± 1.8 ms (27-34) |
| 12:45-12:55 | normal, new process | 14 | 0.17 ± 0.01 ms | 32.1 ± 1.1 ms (31-34) |

Impact → S! from the server log: 0.3 ± 3.6 ms (n=44) normal, −0.4 ± 2.9 ms (n=12) real-time, dominated by the impact estimate. Verdict: do not make real-time priority a default.

# Setup

| | |
|---|---|
| OpenFlight | `feat/iwr-self-trigger` + 2 local commits (OPS idle, backlight) |
| IWR6843 firmware | 1.0.3 (git 4e4e9c5d94b7, hybrid-cadence, built 2026-10-09) |
| OPS243-A | firmware 1.3.3, GPIO UART at 230400 baud, rolling buffer S#24, 30 kS/s |
| Trigger | IWR self-trigger `triggerCfg 36 <snr> 1`, default offset 0.2 m, no sound sensor |
| Geometry | ball 1.57 m, net 5.30 m, IWR board centre 0.04 m above the floor, tilt 10.1° |
| Camera | OV9281, 320×200 at 450 fps (430.9 delivered, 0 gaps on 127 captures) |
| Reference | TrackMan 4, exported with SwingSync |
| Storage | ~6.6 MB per trigger with camera and `--debug`; 0.83 GB for this session |






# Next studio session

- **Before the kiosk:** sweep the self-trigger with `scripts/iwr6843/swing_trigger.py` at SNR 1.5, 2.0 and 3.0. It measures the lane's noise floor, arms the trigger and prints PASS or FAIL per swing, with no OPS and no 7.7 s blind window. Count fires for real swings, addresses, waggles, takeaways and walking past.
- **Robinante's log tails** in two extra SSH terminals for the whole session (forgotten this time).
- **Self-trigger flags**, one block each: `--iwr6843-self-trigger-offset-m` 0.4 and 0.6, `--iwr6843-ball-detector` on and follow (drives the “ball placed” banner), `--iwr6843-veto-no-ball`, then `self-trigger-bin`, SNR and `ball-snr`.
- **Watch the J3 window live** for the ×1.49 switch (34 → 48 ms) and query the OPS before restarting when it happens.
- **Robinante's offset test** against TrackMan interference: `?F`, `T=-2`, wait 15 s, `?F`, with OpenFlight stopped.
- **A block without `--debug`** to shorten the blind window, and SwingSync export again for `python -m openflight_bench.session compare`.

# Logs

Every text file from the session: the kiosk logs (every IWR trigger, OPS verdict and shot, false ones included), the OPS event logs, the session JSONL (shots, OPS captures with raw I/Q, IWR results), the J3 edge log, the camera trigger stamps, the TrackMan export, the notes taken during the session, and each run's launcher.

| FILE | SIZE |
|---|---|
| camera_triggers_20261101.csv | 10 KB |
| captures_20261101.json | 46 KB |
| kiosk_20261101_111546.log | 87 KB |
| kiosk_20261101_113221.log | 355 KB |
| kiosk_20261101_123514.log | 82 KB |
| kiosk_20261101_124549.log | 80 KB |
| launcher/run-openflight.sh | 3 KB |
| launcher/run-openflight.sh.bak-20261101-103654 | 2 KB |
| launcher/run-openflight.sh.bak-20261101-111430 | 2 KB |
| launcher/run-openflight.sh.bak-20261101-112916 | 2 KB |
| launcher/run-openflight.sh.bak-20261101-113135 | 3 KB |
| launcher/run-openflight.sh.bak-20261101-123222 | 3 KB |
| launcher/run-openflight.sh.bak-20261101-124545 | 3 KB |
| lcmf_height_replay.txt | 4 KB |
| ops_j3_20261101_111438.log | 6 KB |
| radar_raw_20261101_111554.log | 7 KB |
| radar_raw_20261101_113228.log | 30 KB |
| radar_raw_20261101_123520.log | 6 KB |
| radar_raw_20261101_124555.log | 6 KB |
| session_20261101_111554_trackman.jsonl | 621 KB |
| session_20261101_113228_trackman.jsonl | 2,960 KB |
| session_20261101_123520_trackman.jsonl | 669 KB |
| session_20261101_124555_trackman.jsonl | 740 KB |

| FILE | SIZE |
| --- | --- |
| swingsync-2026-11-01.csv | 20 KB |
| trackman-build-info.txt | 1 KB |
| trackman-local-changes.patch | 1 KB |
| trackman-notes.txt | 16 KB |
| trackman_compare.txt | 9 KB |

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Data, tools and the full report: HuggeK/OpenFlight-Radar-Workbench, `docs/reports/studio-session-2026-11-01`. Numbers come from the kiosk logs, the J3 edge log, the camera metadata and the SwingSync export, through `python -m openflight_bench.session`.